IGCSE 0607 Practice Questions — E5.6 — Combined Questions & Markscheme

Circle theorems I

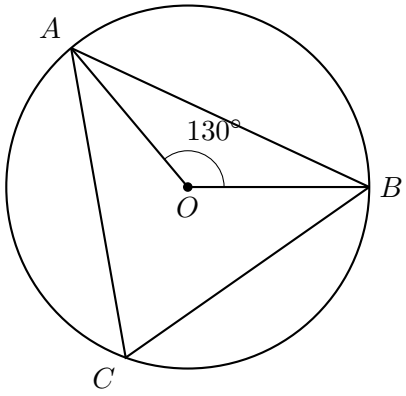
Total Questions: 45 — **Total Marks:** 253 Marks

Table of Contents & Quick Navigation

- **E5.6** Circle theorems I
  → **Paper 2 (Non-Calculator)** *(24 Qs)* · **Paper 4 (Calculator)** *(21 Qs)*

---

1 Points A, B and C lie on a circle, centre O.
Angle $AOB = 130°$.



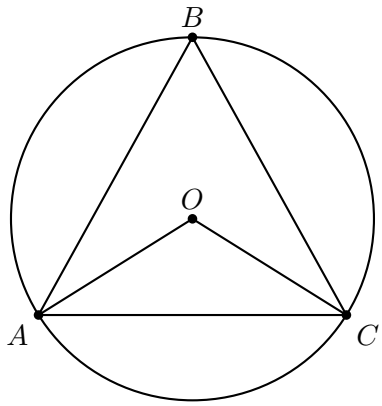
(a) Find the size of angle ACB.

..... ° **[1]**

(b) Find the size of angle OAB.

..... ° **[2]**

2 A, B and C are points on a circle, center O.
Angle $ABC = 58°$.



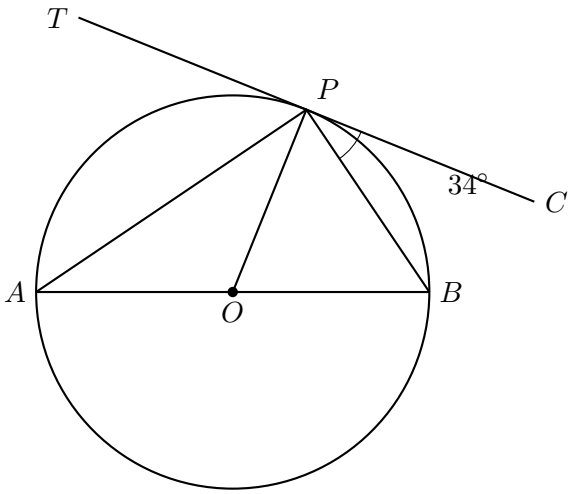
(a) Work out the obtuse angle AOC.

Angle $AOC =$ ..... ° **[1]**

(b) Work out angle OAC.

Angle $OAC =$ ..... ° **[2]**

3 The diagram shows a circle, centre O.
A, B and P are points on the circumference.
AB is a diameter of the circle.
The line TC is a tangent to the circle at the point P.
Angle $BPC = 34°$.



NOT TO SCALE

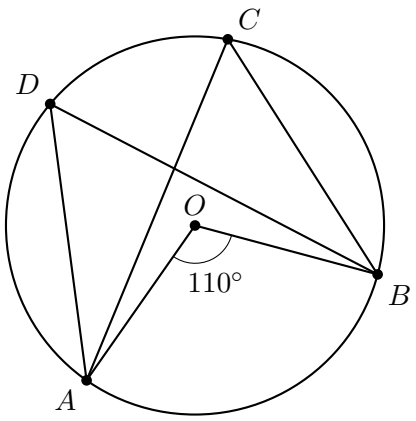
(a) Write down the value of angle PAB.

Angle $PAB =$ ..... ° **[1]**

(b) Work out the value of angle POB.

Angle $POB =$ ..... ° **[2]**

4 A, B, C and D are points on a circle, centre O.
Angle $AOB = 110°$.



NOT TO SCALE

(a) Find the size of angle ACB.

Angle $ACB =$ ..... ° **[1]**

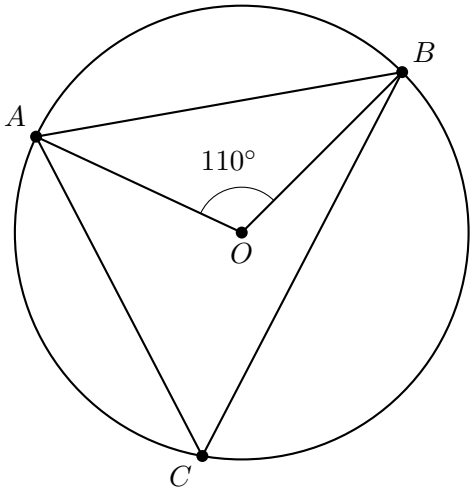
(b) Find the size of angle ADB.

Angle $ADB =$ ..... ° **[1]**

(c) State the geometrical reason for your answer to part (b).

..... **[1]**

5 The diagram shows a circle, centre O.
A, B and C are points on the circumference.
Angle $AOB = 110°$.



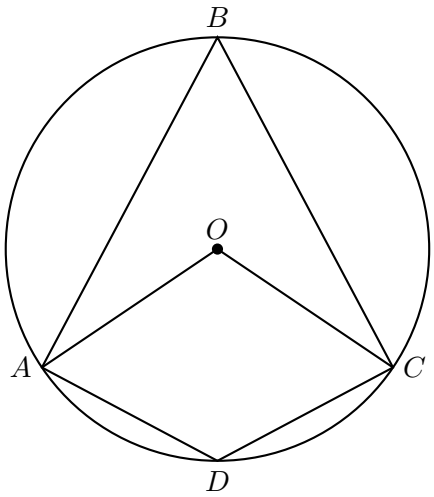
(a) **(i)** Calculate the size of angle ACB.

Angle $ACB =$ ..... ° **[1]**

(ii) Calculate the size of angle OAB.

Angle $OAB =$ ..... ° **[2]**

6 A, B, C and D are points on a circle, center O.
Angle $ABC = 56°$.



(a) Calculate the obtuse angle AOC.

..... ° **[1]**

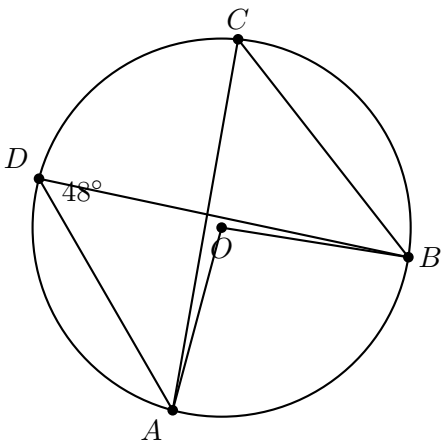
(b) Calculate angle ADC.

..... ° **[1]**

(c) State the geometric reason for your answer to part (b).

..... **[1]**

7 A, B, C and D are points on a circle, centre O.
Angle $ADB = 48°$.



(a) **(i)** Find angle ACB.

Angle $ACB =$ ..... ° **[1]**

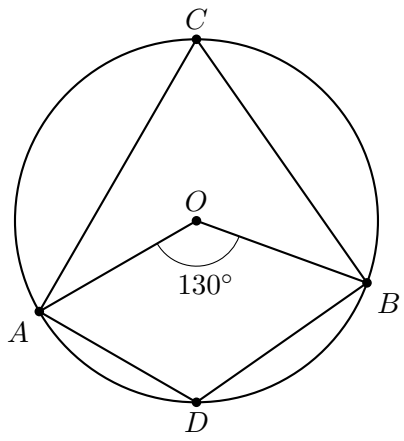
(ii) Give a geometrical reason for your answer to part (a)(i).

..... **[1]**

(b) Find angle AOB.

Angle $AOB =$ ..... ° **[1]**

8 Points A, B, C and D lie on the circumference of a circle, center O.
Angle $AOB = 130°$.



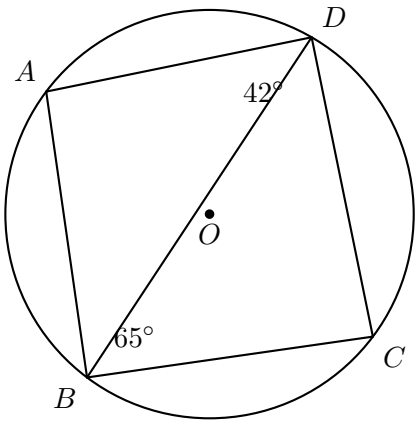
(a) Calculate angle ACB.

..... ° **[1]**

(b) Calculate angle ADB.

..... ° **[2]**

9 A, B, C and D are points on the circumference of a circle, centre O.
Angle $BDC = 42°$ and angle $CBD = 65°$.



(a) Find angle BAC.

Angle $BAC =$ ..... ° **[1]**

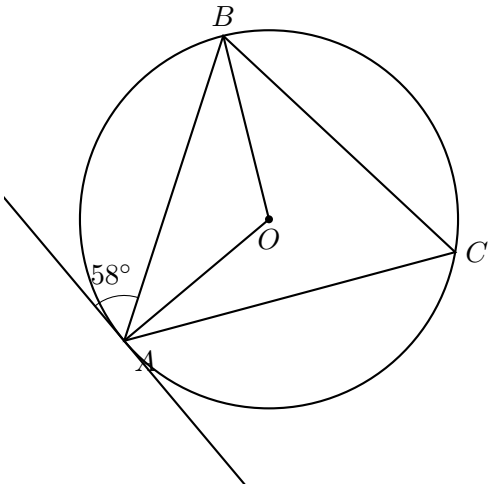
(b) Find angle BCD.

Angle $BCD =$ ..... ° **[1]**

(c) Find angle BAD.

Angle $BAD =$ ..... ° **[2]**

10 The points A, B, and C lie on a circle, centre O.
The line SAT is a tangent to the circle at A.
Angle $SAB = 58°$.



(a) Find angle ACB and state a geometrical reason for your answer.

(i) Angle ACB

..... ° **[1]**

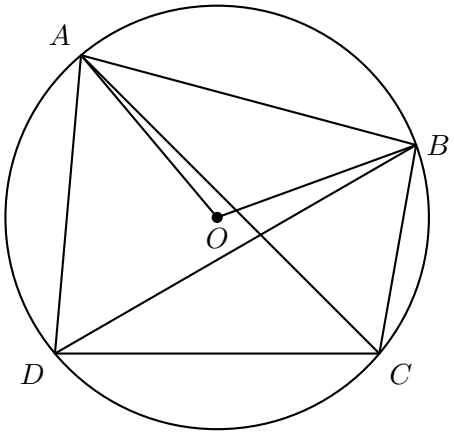
(ii) Reason

..... **[1]**

(b) Find angle OBA.

..... ° **[2]**

11 The diagram shows a circle with centre O.
The points A, B, C and D lie on the circumference of the circle.
Angle $AOB = 110°$ and angle $BDC = 30°$.



(a) Calculate angle ACB.

..... ° **[1]**

(b) State the geometrical reason for your answer to part (a).

..... **[1]**

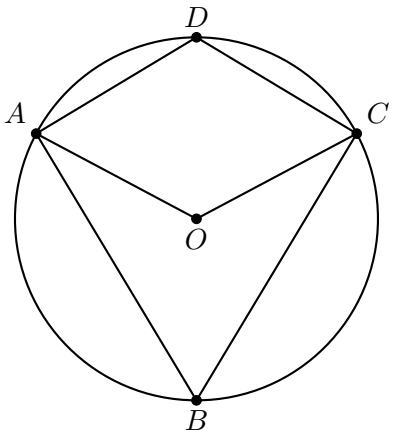
(c) Calculate angle BAC.

..... ° **[1]**

(d) Calculate angle OAB.

..... ° **[2]**

12 A, B, C and D are points on the circumference of a circle, centre O.
Angle $AOC = 124°$.



(a) Calculate angle ABC.

..... ° **[2]**

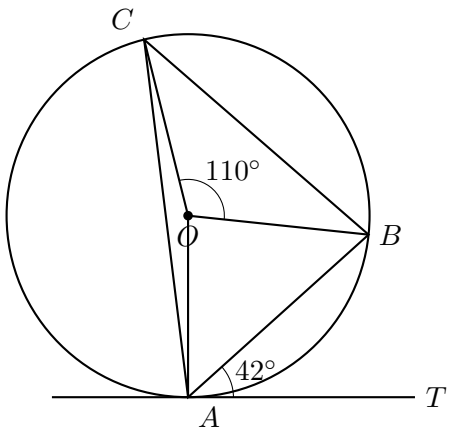
(b) Calculate angle ADC.

..... ° **[2]**

(c) Calculate angle OAC.

..... ° **[2]**

13 A, B and C are points on a circle, centre O.
TA is a tangent to the circle at A.
C is a point on the major arc AB.
Angle $TAB = 42°$ and angle $BOC = 110°$.



(a) Write down the value of

(i) angle BCA,

Angle $BCA =$ ..... ° **[1]**

(ii) angle BAC.

Angle $BAC =$ ..... ° **[1]**

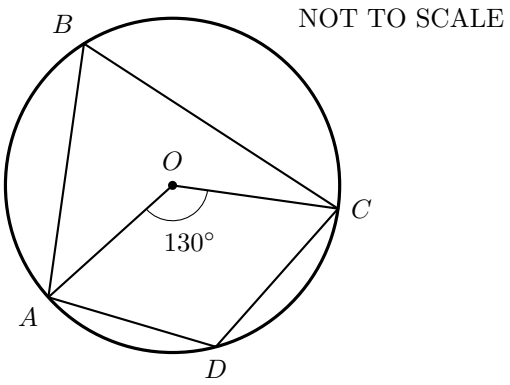
(b) Calculate angle OAB.

Angle $OAB =$ ..... ° **[2]**

(c) Calculate angle ABC.

Angle $ABC =$ ..... ° **[2]**

14 The diagram shows a circle, centre O, with points A, B, C, and D lying on the circumference.
Angle $AOC = 130°$ and angle $OAB = 22°$.



(a) Find the size of angle ABC and give a geometrical reason for your answer.

Angle $ABC =$ ..... , Reason: ..... **[2]**

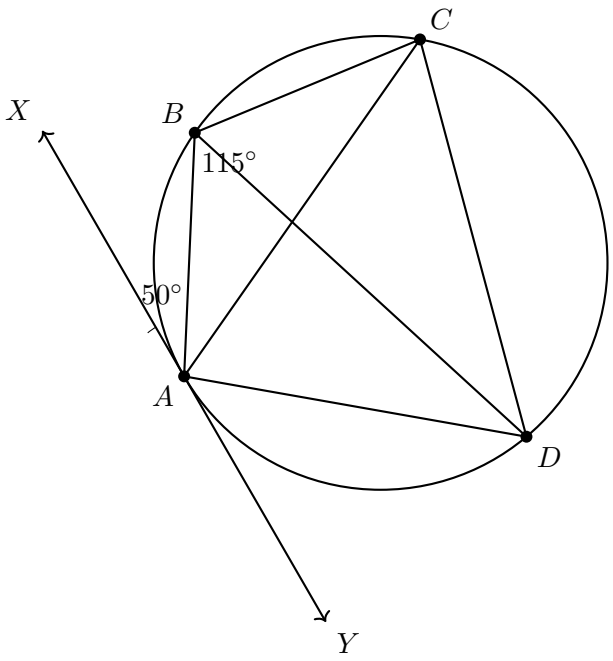
(b) Find the size of angle ADC and give a geometrical reason for your answer.

Angle $ADC =$ ..... , Reason: ..... **[2]**

(c) Calculate the size of angle OCB.

Angle $OCB =$ ..... ° **[2]**

15 The diagram shows a cyclic quadrilateral $ABCD$.
The line XY is a tangent to the circle at A.
Angle $XAD = 50°$ and angle $ABC = 115°$.



(a) **(i)** Write down the size of angle ACD.

Angle $ACD =$ ..... ° **[1]**

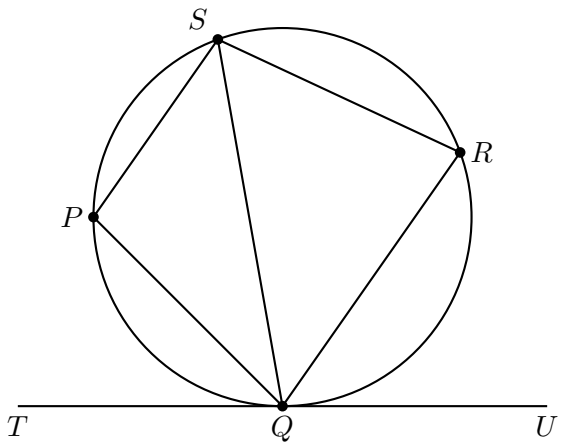
(ii) Calculate the size of angle ADC.

Angle $ADC =$ ..... ° **[1]**

(iii) Calculate the size of angle CAD.

Angle $CAD =$ ..... ° **[2]**

16 P, Q, R and S are points on a circle.
TQU is a tangent to the circle at Q.
Angle $RQU = 55°$ and Angle $QPS = 100°$.



NOT TO SCALE

(a) Find the size of angle QSR.

Angle $QSR =$ ..... ° **[1]**

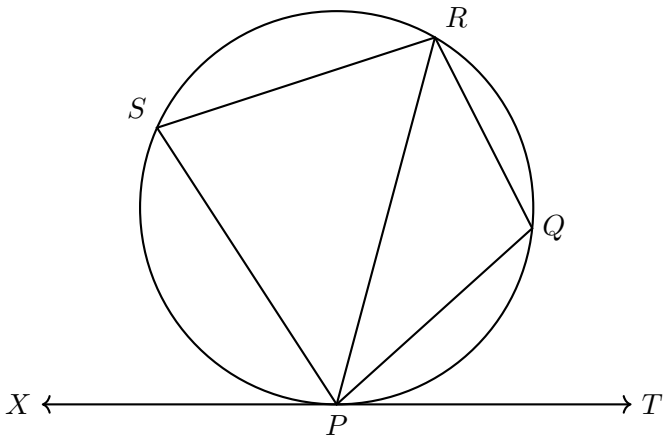
(b) Find the size of angle QRS.

Angle $QRS =$ ..... ° **[2]**

(c) Calculate the size of angle SQR.

Angle $SQR =$ ..... ° **[2]**

17 Points P, Q, R and S lie on a circle.
The straight line XPT is a tangent to the circle at P.
Angle $QPT = 42°$, angle $PQR = 105°$ and angle $SPR = 48°$.



(a) Find the size of angle PRQ.

..... ° **[1]**

(b) Find the size of angle PSR.

..... ° **[1]**

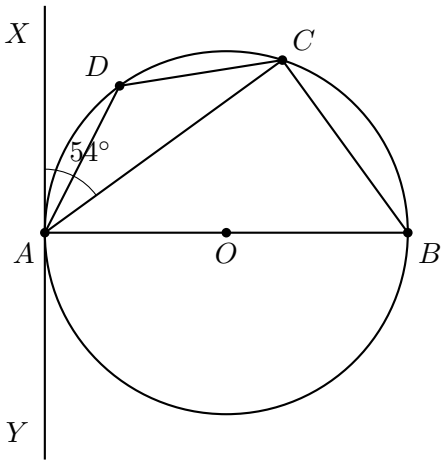
(c) Find the size of angle PRS.

..... ° **[2]**

(d) Find the size of angle XPS.

..... ° **[1]**

18 In the diagram, AB is a diameter of the circle, center O.
The line XAY is a tangent to the circle at A.
C and D are points on the circle.
Angle $XAC = 54°$.



(a) **(i)** Write down the size of angle ACB.

..... ° **[1]**

(ii) Give a geometric reason for your answer.

..... **[1]**

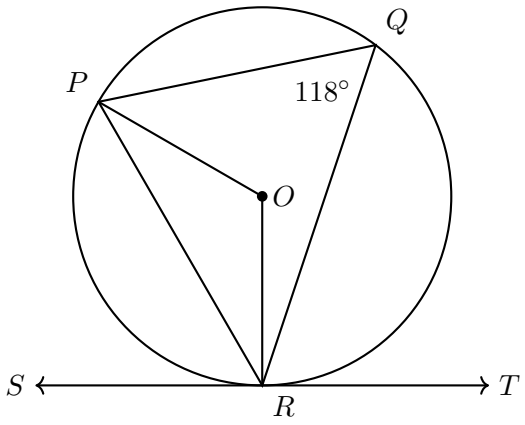
(b) Find angle ABC.

..... ° **[1]**

(c) Find angle ADC.

..... ° **[2]**

19 P, Q and R are points on a circle, centre O.
ST is a tangent to the circle at R.
Angle $PQR = 118°$.



(a) Find reflex angle POR.

Reflex angle $POR =$ ..... ° **[2]**

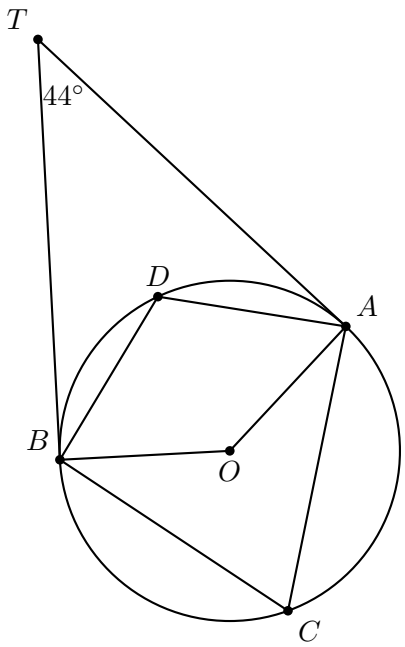
(b) Find obtuse angle POR.

Obtuse angle $POR =$ ..... ° **[1]**

(c) Find angle PRS.

Angle $PRS =$ ..... ° **[2]**

20 A, B, C and D are points on a circle, centre O.
TA and TB are tangents to the circle at A and B.
Angle $ATB = 44°$.



(a) Calculate angle AOB.

Angle AOB = ... ° **[2]**

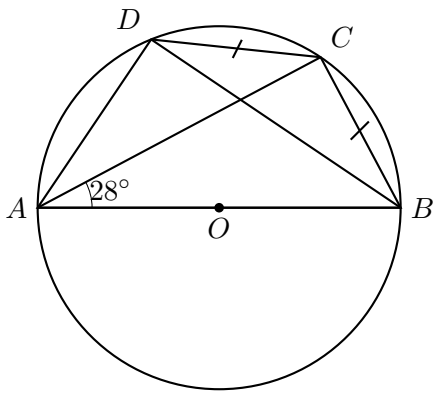
(b) Calculate angle ACB.

Angle ACB = ... ° **[1]**

(c) Calculate angle ADB.

Angle ADB = ... ° **[2]**

21 A, B, C and D are points on a circle.
AB is a diameter of the circle.
$CD = CB$ and angle $CAB = 28°$.



(a) (i) Find angle CBA.

Angle $CBA =$ ..... ° **[1]**

(ii) Give a geometrical reason for your answer.

..... **[1]**

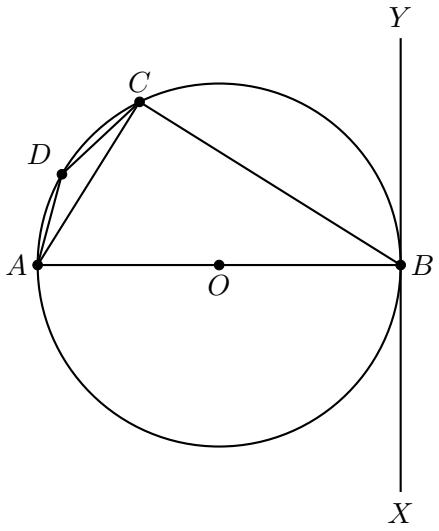
(b) Find angle CAD.

Angle $CAD =$ ..... ° **[1]**

(c) Find angle ADC.

Angle $ADC =$ ..... ° **[2]**

22 A, B, C and D lie on a circle, centre O.
AB is a diameter of the circle.
The line XY is a tangent to the circle at B.
Angle $CBY = 58°$.



(a) (i) Find angle BAC.

..... ° **[1]**

(ii) State the circle theorem that gives this angle directly from angle CBY.

..... **[1]**

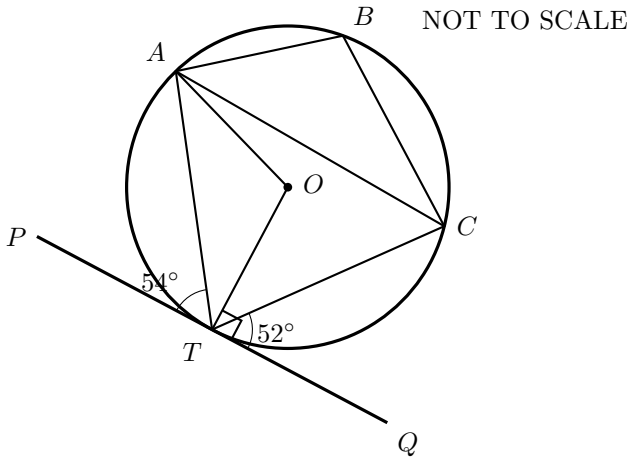
(b) Calculate angle ABC.

..... ° **[3]**

(c) Calculate angle ADC.

..... ° **[2]**

23 The diagram shows a circle, centre O, with points A, B, C, and T lying on the circumference.
The straight line PTQ is a tangent to the circle at T.
Angle $PTA = 54°$, angle $QTC = 52°$, and angle $BCT = 86°$.



(a) Find the size of angle TAC and give a geometrical reason for your answer.

Angle $TAC =$ ..... , Reason: ..... **[2]**

(b) Calculate the size of

(i) angle ABC,

Angle $ABC =$ ..... ° **[2]**

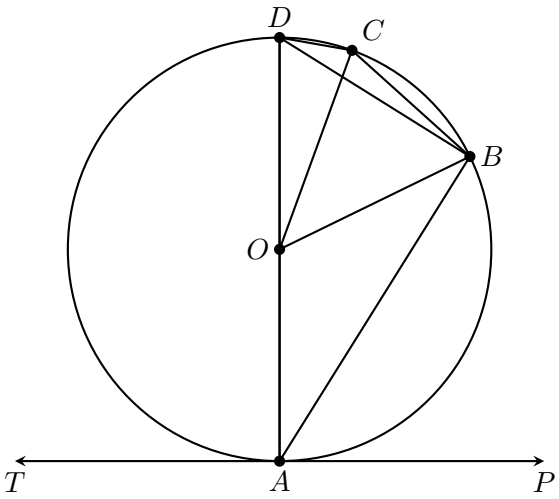
(ii) angle ACB,

Angle $ACB =$ ..... ° **[2]**

(iii) angle OAT.

Angle $OAT =$ ..... $^\circ$ **[3]**

24 A, B, C and D are points on a circle, center O.
AD is a diameter of the circle.
PAT is a tangent to the circle at A.
Angle $PAB = 58°$ and angle $BDC = 22°$.



(a) Calculate angle ADB.

Angle ADB = ... ° **[2]**

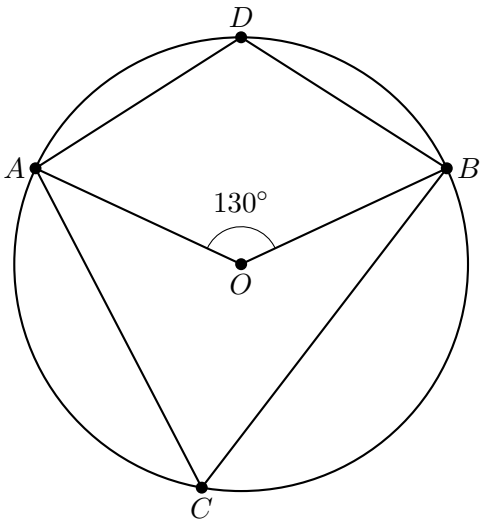
(b) Calculate angle ABC.

Angle ABC = ... ° **[2]**

(c) Calculate angle OBC.

Angle OBC = ... ° **[3]**

25 The points A, B, C, and D lie on a circle with centre O.
Angle $AOB = 130°$.



(a) Calculate angle ACB.

Angle $ACB =$ ..... ° **[1]**

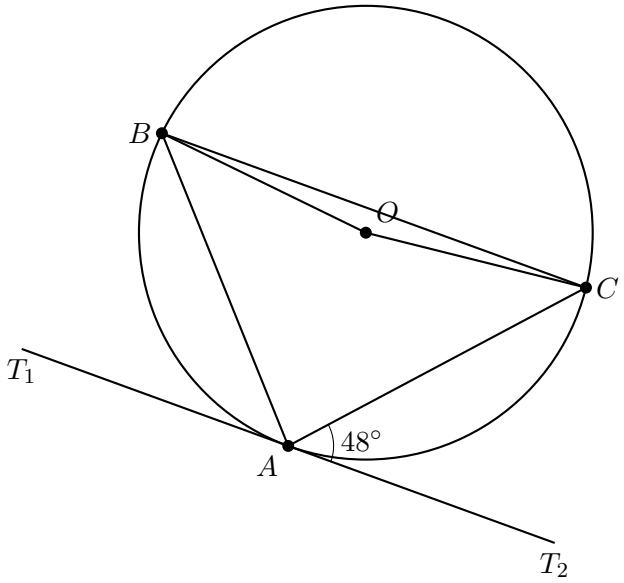
(b) Calculate angle ADB.

Angle $ADB =$ ..... ° **[1]**

(c) Give a geometrical reason for your answer to part (b).

..... **[1]**

26 The points A, B, and C lie on a circle with centre O.
The straight line T_1AT_2 is a tangent to the circle at A.
The chord BC is parallel to the tangent T_1AT_2.
Angle $CAT_2 = 48°$.



(a) Find angle ABC.

Angle ABC = ... ° **[1]**

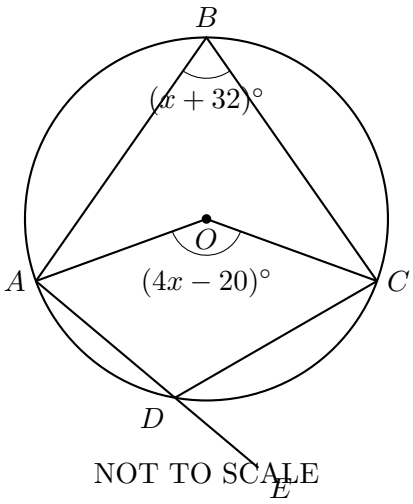
(b) Calculate angle BAC.

Angle BAC = ... ° **[2]**

(c) Calculate angle OBC.

Angle OBC = ... ° **[2]**

27 A, B, C and D lie on a circle, centre O.
Angle $ABC = (x + 32)°$ and the obtuse angle $AOC = (4x - 20)°$.
The line AD is extended to the point E.



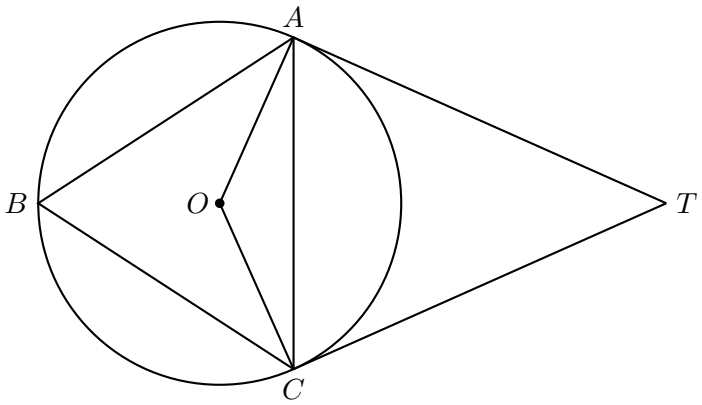
(a) Find the value of x.

$x =$ ..... **[3]**

(b) Work out the size of angle CDE.

Angle $CDE =$ ..... ° **[2]**

28 A, B and C are points on a circle, center O.
The lines TA and TC are tangents to the circle at A and C respectively.
Angle $ATC = 48°$.



(a) Calculate angle AOC.

..... ° **[2]**

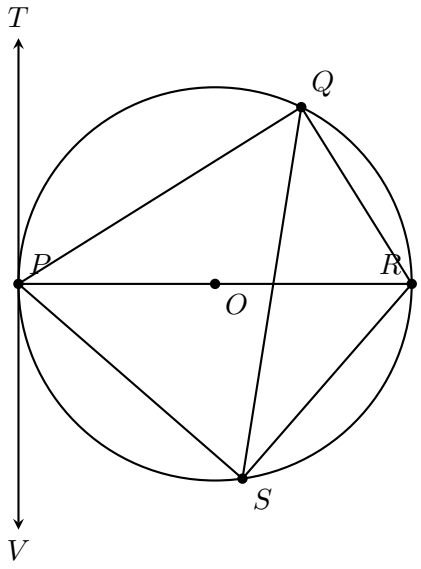
(b) Calculate angle ABC.

..... ° **[1]**

(c) Calculate angle OAC.

..... ° **[2]**

29 P, Q, R and S are points on a circle, center O.
PR is a diameter of the circle.
TPV is a tangent to the circle at P.
Angle $QPR = 32°$ and angle $RPS = 41°$.



(a) Calculate angle PRQ.

Angle $PRQ =$ ..... ° **[2]**

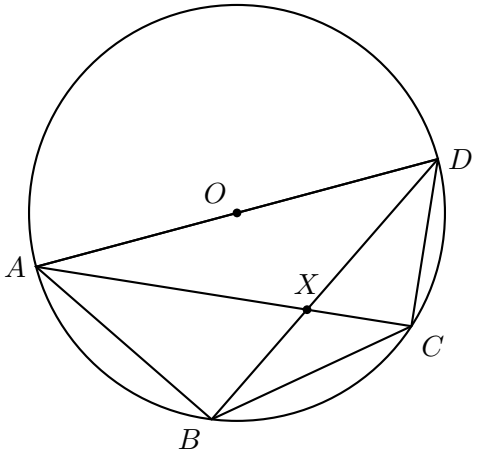
(b) Calculate angle TPQ.

Angle $TPQ =$ ..... ° **[2]**

(c) Calculate angle PQS.

Angle $PQS =$ ..... ° **[2]**

30 AD is a diameter of the circle, centre O.
The points A, B, C, and D lie on the circumference.
The chords AC and BD intersect at X.
Angle $BAC = 32°$ and angle $CAD = 24°$.



(a) Find angle ADC.

..... ° **[2]**

(b) Find angle ABC.

..... ° **[1]**

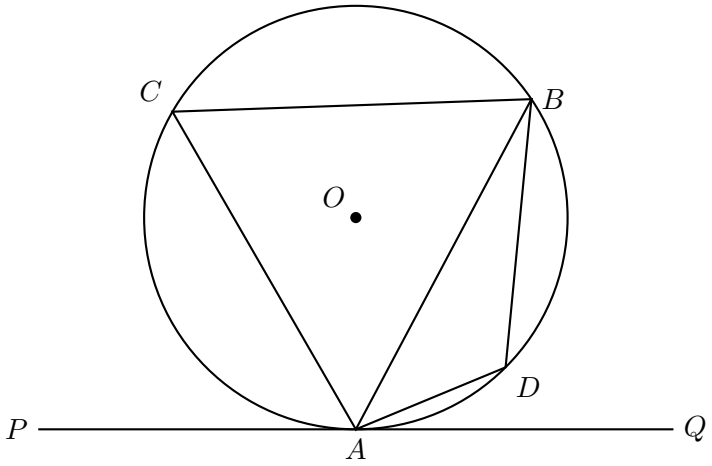
(c) Find angle DBC.

..... ° **[1]**

(d) Find angle AXB.

..... ° **[2]**

31 The diagram shows a circle with centre O.
The line PAQ is a tangent to the circle at the point A.
The points B, C and D lie on the circle.
Angle $QAB = 62°$.



(a) Calculate angle ACB and give a geometrical reason for your answer.

$Angle ACB =$ ..... ° , Reason: ..... **[2]**

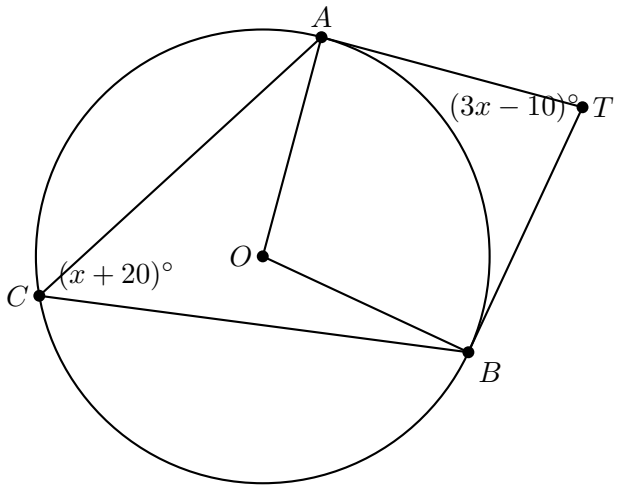
(b) Calculate angle ADB.

..... ° **[2]**

(c) Calculate angle OBA.

..... ° **[2]**

32 The diagram shows a circle, centre O.
A, B and C are points on the circumference.
The lines TA and TB are tangents to the circle at A and B respectively.
Angle $ATB = (3x - 10)°$ and angle $ACB = (x + 20)°$.



(a) Find the value of x.

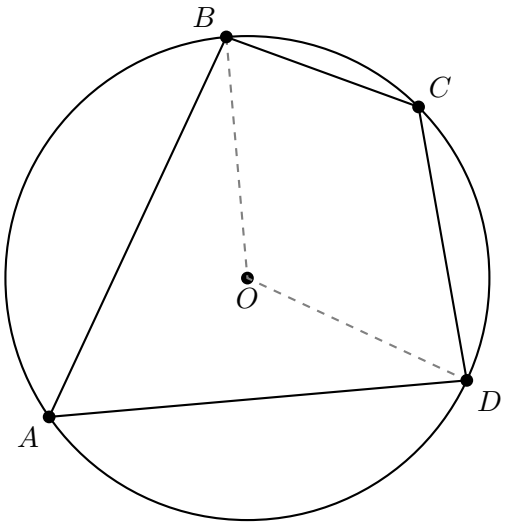
$x =$ ..... **[3]**

(b) Calculate the size of angle OAB.

Angle $OAB =$ ..... ° **[2]**

33 The points A, B, C, and D form a cyclic quadrilateral on a circle with centre O.
The interior angles of the quadrilateral are given in terms of x and y:

$$\angle DAB = (3x + 2y)^\circ$$
$$\angle ABC = (5x + 3y)^\circ$$
$$\angle BCD = (9x + 2y)^\circ$$
$$\angle CDA = (4x + 3y)^\circ$$



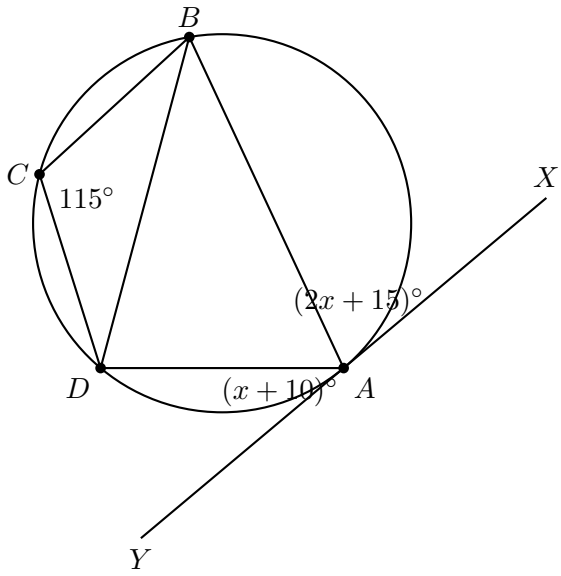
(a) By setting up and solving simultaneous equations, find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(b) Calculate the obtuse angle BOD.

Angle $BOD =$ ..... $^\circ$ **[2]**

34 The points A, B, C and D lie on a circle.
XY is a tangent to the circle at A.
Angle $BCD = 115°$, angle $XAB = (2x + 15)°$ and angle $YAD = (x + 10)°$.



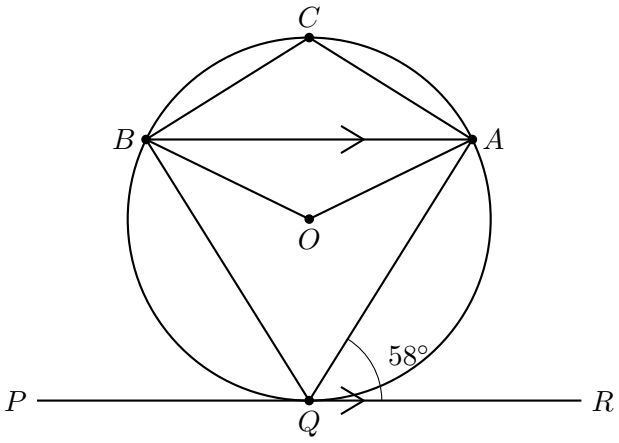
(a) Find the value of x.

$x =$ ..... **[4]**

(b) Calculate angle ABD.

Angle $ABD =$ ..... ° **[2]**

35 A, B, C and Q are points on a circle, center O.
The line PQR is a tangent to the circle at Q.
The chord AB is parallel to PR.
Angle $AQR = 58°$.



(a) **(i)** Find the size of angle ABQ.

..... ° **[1]**

(ii) Give a geometric reason for your answer.

..... **[1]**

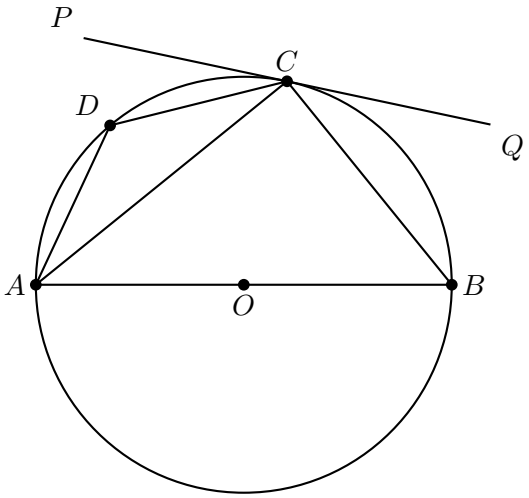
(b) Find angle AOB.

..... ° **[2]**

(c) Find angle ACB.

..... ° **[2]**

36 A, B, C and D are points on a circle, centre O.
AB is a diameter of the circle.
PQ is a tangent to the circle at C.
Angle $CAB = (2x - 5)°$ and Angle $BCQ = (x + 17)°$.



NOT TO SCALE

(a) Set up an equation in terms of x and solve it to show that $x = 22$.
Give a geometrical reason to justify your equation.

[3]

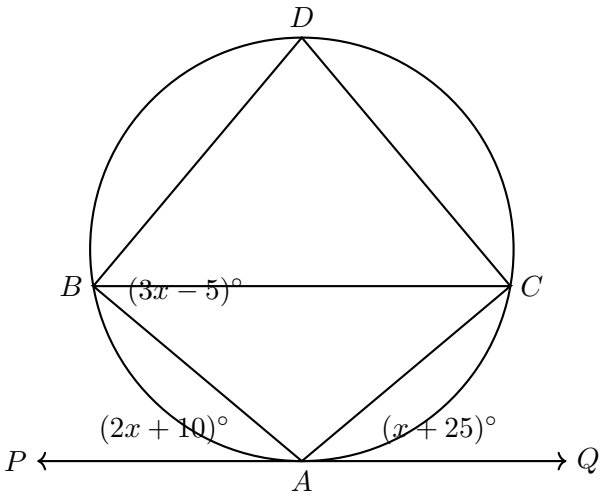
(b) Work out the size of angle CBA.

Angle $CBA =$ ..... ° **[2]**

(c) Find the size of angle ADC.

Angle $ADC =$ ..... ° **[2]**

37 Points A, B, C and D lie on a circle.
The straight line PAQ is a tangent to the circle at A.
Angle $PAB = (2x + 10)°$, angle $ABC = (3x - 5)°$ and angle $CAQ = (x + 25)°$.
D is a point on the major arc BC.



(a) Find the value of x.
Show all your working.

$x =$ ..... **[3]**

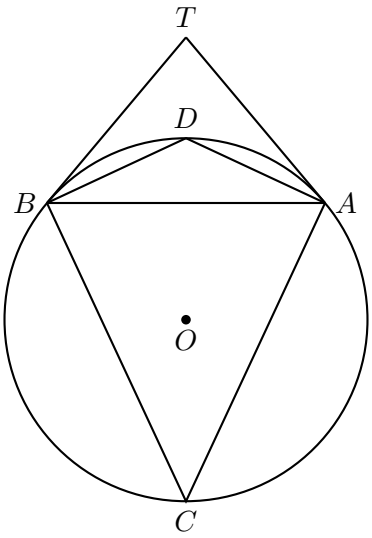
(b) Find the size of angle BAC.

..... ° **[2]**

(c) Find the size of angle BDC.

..... ° **[2]**

38 TA and TB are tangents to a circle, centre O, at the points A and B.
C is a point on the major arc AB and D is a point on the minor arc AB.
Angle $ATB = (x + 10)°$
Angle $TAB = (2y - 10)°$
Angle $ACB = (3y - 40)°$



(a) Find the value of y.

$y =$ ..... **[3]**

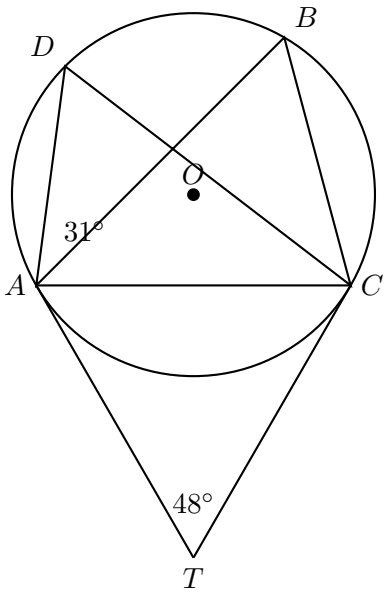
(b) Find the value of x.

$x =$ ..... **[2]**

(c) Find angle ADB.

Angle $ADB =$ ..... ° **[2]**

39 A, B, C and D are points on a circle, centre O.
TA and TC are tangents to the circle at A and C.
Angle $ATC = 48°$ and angle $CAD = 31°$.



(a) **(i)** Find angle ABC.

Angle $ABC =$ ..... ° **[2]**

(ii) Find angle ADC.

Angle $ADC =$ ..... ° **[2]**

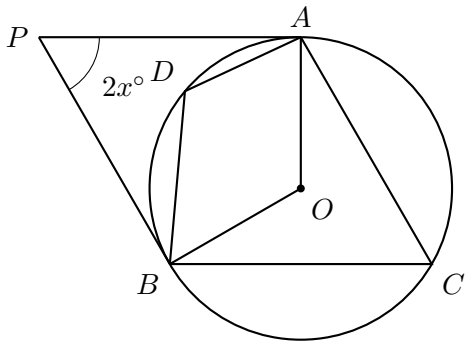
(iii) Find angle ACD.

Angle $ACD =$ ..... ° **[1]**

(iv) Calculate angle AOC.

Angle $AOC =$ ..... ° **[2]**

40 The points A, B, C, and D lie on a circle, centre O.
PA and PB are tangents to the circle at A and B.
Angle $APB = 2x°$.
C is a point on the major arc AB and D is a point on the minor arc AB.



(a) Find an expression, in terms of x, for each of the following angles.
Give each answer in its simplest form.

Angle AOB

..... ° **[1]**

(b) Angle ACB

..... ° **[1]**

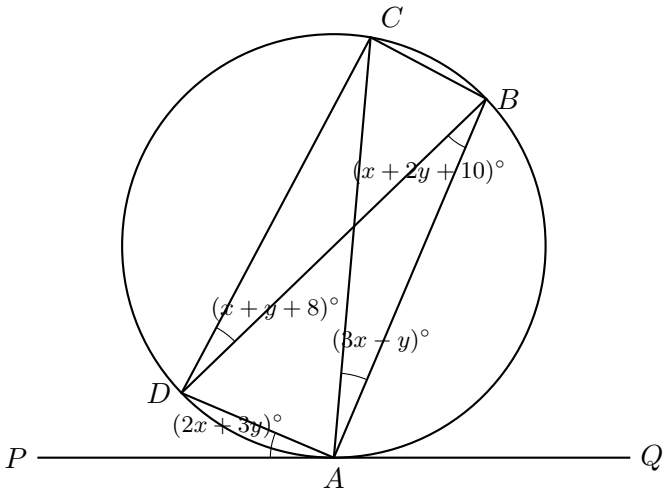
(c) Angle ADB

..... ° **[2]**

(d) Given that angle $ADB = 3 \times$ angle ACB, find the value of x.

$x =$ ..... **[3]**

41 The diagram shows a circle containing the points A, B, C and D.
The straight line PAQ is a tangent to the circle at A.
The chords AC and BD intersect.
Angle $PAD = (2x + 3y)^\circ$ and angle $ABD = (x + 2y + 10)^\circ$.
Angle $BAC = (3x - y)^\circ$ and angle $BDC = (x + y + 8)^\circ$.



NOT TO SCALE

(a) **(i)** Use the circle theorems to write down two equations in x and y.

[2]

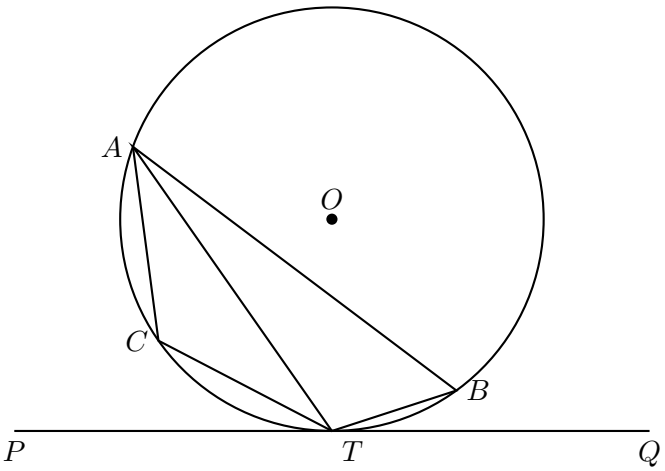
(ii) Solve your equations to find the value of x and the value of y.
Show all your working.

$x =$ ..... , $y =$ ..... **[3]**

(b) Given that angle $ABC = 95^\circ$, work out the size of angle QAB.

Angle $QAB =$ ..... ° **[3]**

42 A, B and C are points on a circle, center O.
The straight line PTQ is a tangent to the circle at the point T.
$\angle ATP = (4x - 5)^\circ$ and $\angle ABT = (2x + 25)^\circ$
$\angle BTQ = (3y)^\circ$ and $\angle BAT = (y + 12)^\circ$



(a) Use the alternate segment theorem to find the values of x and y.

$x =$ ..... , $y =$ ..... **[4]**

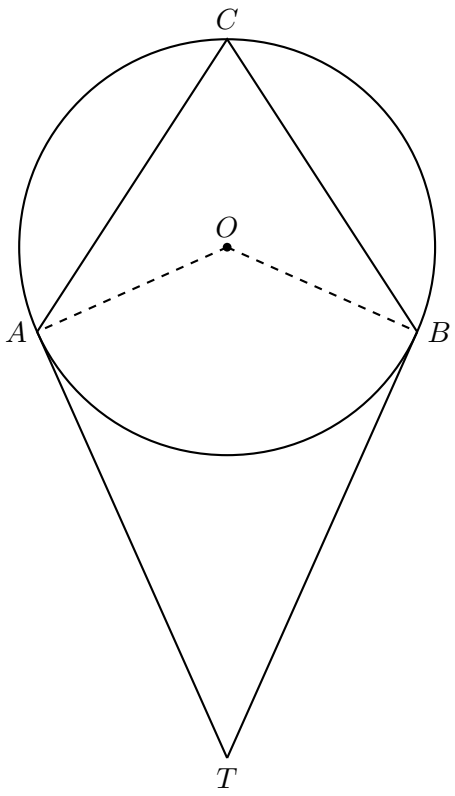
(b) Calculate angle ATB.

..... ° **[2]**

(c) C is a point on the minor arc AT.
Calculate angle ACT.

..... ° **[2]**

43 The diagram shows a circle with centre O.
The lines TA and TB are tangents to the circle at the points A and B.
C is a point on the major arc AB.
Angle $ATB = x°$ and angle $ACB = (2x - 30)°$.



(a) (i) Find the value of x.

$x =$ ..... **[4]**

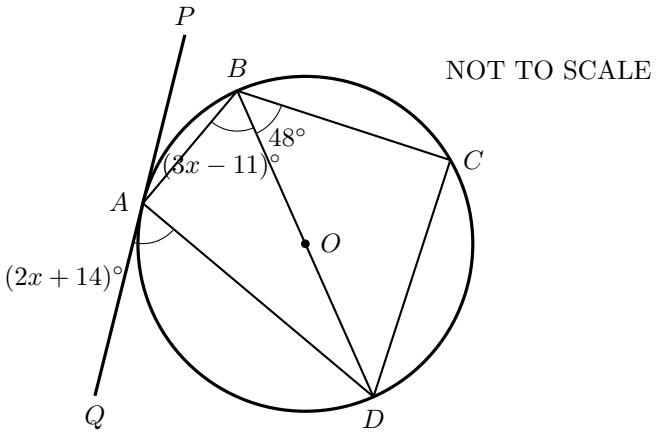
(ii) Calculate angle OAB.

..... ° **[2]**

(b) D is a point on the minor arc AB.
Calculate angle ADB.

..... ° **[2]**

44 The diagram shows a circle, centre O, of radius 9.4 cm. Points A, B, C, and D lie on the circumference, and BD is a diameter of the circle.
The straight line PAQ is a tangent to the circle at A.
Angle $QAD = (2x + 14)°$, angle $ABD = (3x - 11)°$, and angle $CBD = 48°$.



(a) Find the value of x and state the circle theorem you used.

$x =$ ..... , Theorem: ..... **[3]**

(b) Calculate the size of angle ADC, giving a geometrical reason for your answer.

Angle $ADC =$ ..... , Reason: ..... **[2]**

(c) Calculate the length of chord AC, giving your answer correct to 3 significant figures.

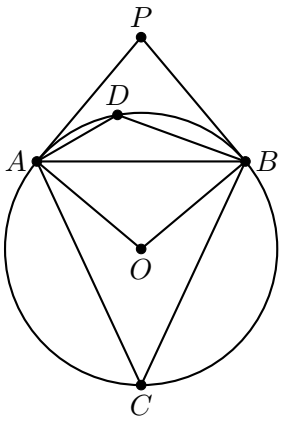
$AC =$ ..... cm **[3]**

(d) Calculate the area of the minor segment bounded by chord AD and the minor arc AD, giving

your answer correct to 3 significant figures.

..... cm^2 **[3]**

45 PA and PB are tangents to a circle, centre O, from a point P.
C is a point on the major arc AB.
Angle $APB = 4x°$.
Angle $ACB = (3x - 10)°$.



(a) Find an expression for angle PAB in terms of x.

..... ° **[3]**

(b) Use the alternate segment theorem to form an equation in x, and show that $x = 20$.

[3]

(c) Calculate angle AOB.

..... ° **[3]**

(d) D is a point on the minor arc AB.
Calculate angle ADB.

..... ° **[3]**

1 **(a)** 65 **[B1]**

(b) Uses isosceles triangle OAB, e.g. $\dfrac{180-130}{2}$ **[M1]**

25 **[A1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** 116 **[B1]**

(b) $\dfrac{180-116}{2}$ **[M1]**

32 **[A1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** 34 **[B1]**

(b) Recognises angle at centre is twice angle at circumference $\implies 2 \times 34$ **[M1]**

68 **[A1]**

[Back to Question] *[Total Marks: 3]*

4 **(a)** 55 **[B1]**

(b) 55 **[B1]**

(c) Angles in the same segment (are equal) **[B1]**

[Back to Question] *[Total Marks: 3]*

5 **(a)** **(i)** 55 **[B1]**

(ii) Recognising triangle OAB is isosceles (radii) OR $(180-110) \div 2$ **[M1]**

35 **[A1]**

[Back to Question] *[Total Marks: 3]*

6 **(a)** 112 **[B1]**

(b) 124 **[B1]**

(c) Opposite angles of a cyclic quadrilateral sum to 180° **[B1]**

[Back to Question] *[Total Marks: 3]*

7 **(a)** **(i)** 48 **[B1]**

(ii) Angles in the same segment **[B1]**

(b) 96 **[B1]**

[Back to Question] *[Total Marks: 3]*

8 **(a)** 65 **[B1]**

(b) $180 - 65$ **[M1]**

115 **[A1]**

[Back to Question] *[Total Marks: 3]*

9 **(a)** 42 **[B1]**

(b) $180 - (42 + 65) = 73$ **[B1]**

(c) Angle BAD + Angle $BCD = 180$ (cyclic quadrilateral) **[M1]**

$180 - 73 = 107$ **[A1]**

[Back to Question] *[Total Marks: 4]*

10 **(a)** **(i)** 58 **[B1]**

(ii) Alternate segment theorem **[B1]**

(b) Angle $AOB = 2 \times 58 = 116°$ **[M1]**

Angle $OBA = \dfrac{180 - 116}{2} = 32°$ **[A1]**

[Back to Question] *[Total Marks: 4]*

11 **(a)** 55 **[B1]**

(b) Angle at the centre is twice the angle at the circumference. **[B1]**

(c) 30 **[B1]**

(d) $(180 - 110) \div 2$ **[M1]**

35 **[A1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** Angle at centre is twice the angle at the circumference **[M1]**

62 **[A1]**

(b) Opposite angles in a cyclic quadrilateral sum to 180° **[M1]**

$180 - 62 = 118$ **[A1]**

(c) Triangle OAC is isosceles ($OA = OC$) **[M1]**

$(180 - 124) \div 2 = 28$ **[A1]**

[Back to Question] *[Total Marks: 6]*

13 **(a)** **(i)** 42 **[B1]**

(ii) 55 **[B1]**

(b) $90 - 42$ **[M1]**

48 **[A1]**

(c) $180 - (55 + 42)$ **[M1]**

83 **[A1]**

[Back to Question] *[Total Marks: 6]*

14 **(a)** Angle $ABC = 65°$ **[B1]**

Reason: the angle at the centre is twice the angle at the circumference **[B1]**

(b) Angle $ADC = 115°$ **[B1]**

Reason: opposite angles of a cyclic quadrilateral sum to 180° **[B1]**

(c) Angle $OBA = 22°$ ($OA = OB$), so angle $OBC = 65° - 22° = 43°$ **[M1]**

Angle $OCB = 43$ ($OB = OC$) **[A1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** **(i)** 50 **[B1]**

(ii) $180 - 115 = 65$ **[B1]**

(iii) $180 - (50 + 65)$ **[M1]**

65 **[A1]**

[Back to Question] *[Total Marks: 4]*

16 **(a)** 55 **[B1]**

(b) Uses opposite angles of a cyclic quadrilateral (e.g., $180 - 100$) **[M1]**

80 **[A1]**

(c) Uses angles in a triangle sum to 180° (e.g., $180 - (55 + 80)$) **[M1]**

45 **[A1]**

[Back to Question] *[Total Marks: 5]*

17 **(a)** 42 **[B1]**

(b) 75 **[B1]**

(c) Uses sum of angles in triangle PRS equal to 180° ($180 - 75 - 48$) **[M1]**

57 **[A1]**

(d) 57 **[B1]**

[Back to Question] *[Total Marks: 5]*

18 **(a)** **(i)** 90 **[B1]**

(ii) Angle in a semicircle is 90° **[B1]**

(b) 54 **[B1]**

(c) $180 - 54$ **[M1]**

126 **[A1]**

[Back to Question] *[Total Marks: 5]*

19 **(a)** Angle at centre = 2× angle at circumference **[M1]**

$118 \times 2 = 236$ **[A1]**

(b) $360 - 236 = 124$ **[B1]**

(c) Triangle POR is isosceles, so angle OPR = angle $ORP = \dfrac{180 - 124}{2} = 28°$ **[M1]**

Angle $ORS = 90°$ (tangent-radius), so angle $PRS = 90 - 28 = 62$ **[A1]**

[Back to Question] *[Total Marks: 5]*

20 **(a)** Angle $OAT = 90°$ and Angle $OBT = 90°$ **[B1]**

$360 - 90 - 90 - 44 = 136$ **[A1]**

(b) 68 **[B1]**

(c) $180 - 68$ **[M1]**

112 **[A1]**

[Back to Question] *[Total Marks: 5]*

21 **(a)** **(i)** 62 **[B1]**

(ii) Angle in a semicircle (is 90°) **[B1]**

(b) 28 **[B1]**

(c) Angle $ABC = 62°$ OR Angle $ADB = 90°$ **[M1]**

118 **[A1]**

[Back to Question] *[Total Marks: 5]*

22 **(a)** **(i)** 58 **[B1]**

(ii) Alternate segment theorem **[B1]**

(b) Angle in a semicircle is 90° (Angle $ACB = 90°$) **[M1]**

$180 - 90 - 58$ **[M1]**

32 **[A1]**

(c) $180 - 32$ **[M1]**

148 **[A1]**

[Back to Question] *[Total Marks: 7]*

23 **(a)** Angle $TAC = 52°$ **[B1]**

Reason: alternate segment theorem **[B1]**

(b) **(i)** Angle $ATC = 180° - 54° - 52° = 74°$ **[M1]**

Angle $ABC = 74$ (angles in the same segment) **[A1]**

(ii) Angle $ACT = 54°$ (alternate segment theorem) **[M1]**

Angle $ACB = 86° - 54° = 32$ **[A1]**

(iii) Angle $OTP = 90°$ (tangent to radius $= 90°$) **[B1]**

Angle $OTA = 90° - 54° = 36°$ **[M1]**

Angle $OAT = 36$ ($OA = OT$) **[A1]**

[Back to Question] *[Total Marks: 9]*

24 **(a)** Angle $PAD = 90°$ OR Angle $DAB = 90 - 58 = 32°$ **[M1]**

58 **[A1]**

(b) Angle $ADC = 58 + 22 = 80°$ **[M1]**

$180 - 80 = 100$ **[A1]**

(c) Angle $BOC = 2 \times 22 = 44°$ **[M1]**

$\frac{180 - 44}{2}$ **[M1]**

68 **[A1]**

[Back to Question] *[Total Marks: 7]*

25 **(a)** 65 **[B1]**

(b) 115 **[B1]**

(c) Opposite angles of a cyclic quadrilateral sum to 180°. **[B1]**

[Back to Question] *[Total Marks: 3]*

26 **(a)** 48 **[B1]**

(b) Angle $BCA = 48°$ (Alternate interior angles, since $BC \parallel T_1T_2$) **[M1]**

$180 - 48 - 48 = 84°$ **[A1]**

(c) Angle $BOC = 2 \times 84 = 168°$ **[M1]**

Triangle OBC is isosceles ($OB = OC$), so Angle $OBC = (180 - 168) \div 2 = 6°$ **[A1]**

[Back to Question] *[Total Marks: 5]*

27 **(a)** Forms correct equation using angle at centre theorem: $4x - 20 = 2(x + 32)$ **[M1]**

$4x - 20 = 2x + 64 \implies 2x = 84$ **[M1]**

$x = 42$ **[A1]**

(b) Recognises angle CDE = angle ABC (exterior angle of cyclic quadrilateral) OR finds angle $ADC = 180 - (42 + 32) = 106°$ **[M1]**

74 **[A1]**

[Back to Question] *[Total Marks: 5]*

28 **(a)** Recognition that tangent meets radius at 90°, so $\angle OAT = \angle OCT = 90°$ **[M1]**

$180 - 48 = 132$ **[A1]**

(b) 66 **[B1]**

(c) Use of isosceles triangle OAC ($OA = OC$ radii): $(180 - 132) \div 2$ **[M1]**

OR $\angle TAC = (180 - 48) \div 2 = 66°$ leading to $\angle OAC = 90 - 66$ **[M1]**

24 **[A1]**

[Back to Question] *[Total Marks: 5]*

29 **(a)** Angle $PQR = 90°$ **[M1]**

$180 - 90 - 32 = 58$ **[A1]**

(b) Angle $TPR = 90°$ **[M1]**

$90 - 32 = 58$ **[A1]**

(c) Angle $PRS = 180 - 90 - 41 = 49°$ **[M1]**

49 **[A1]**

[Back to Question] *[Total Marks: 6]*

30 **(a)** Angle $ACD = 90°$ (angle in a semicircle) **[M1]**

Angle $ADC = 180 - 90 - 24 = 66°$ **[A1]**

(b) $180 - 66 = 114°$ (opposite angles of cyclic quadrilateral) **[B1]**

(c) 24° (angles in same segment as angle CAD) **[B1]**

(d) Angle $ABD = 90°$ (angle in a semicircle) **[M1]**

In triangle ABX, angle $AXB = 180 - 90 - 32 = 58°$ **[A1]**

[Back to Question] *[Total Marks: 6]*

31 **(a)** Angle $ACB = 62$ **[B1]**

Alternate segment theorem **[B1]**

(b) $180 - 62$ **[M1]**

118 **[A1]**

(c) Angle $OAB = 90 - 62 = 28°$ **[M1]**

28 (since triangle OAB is isosceles) **[A1]**

[Back to Question] *[Total Marks: 6]*

32 **(a)** Angle $AOB = 2(x + 20) = 2x + 40$ **[M1]**

Angle AOB + Angle $ATB = 180° \implies (2x + 40) + (3x - 10) = 180$ **[M1]**

$5x + 30 = 180 \implies 5x = 150 \implies x = 30$ **[A1]**

(b) Angle $AOB = 2(30) + 40 = 100°$ **[M1]**

Angle $OAB = (180 - 100) \div 2 = 40$ **[A1]**

[Back to Question] *[Total Marks: 5]*

33 **(a)** Sets up equation for opposite angles:
$(3x + 2y) + (9x + 2y) = 180 \implies 12x + 4y = 180 \implies 3x + y = 45$ **[M1]**

Sets up second equation:
$(5x + 3y) + (4x + 3y) = 180 \implies 9x + 6y = 180 \implies 3x + 2y = 60$ **[M1]**

Correct method to solve linear simultaneous equations (e.g. subtracting first from second to get $y = 15$) **[M1]**

$x = 10$, $y = 15$ **[A1]**

(b) Finds $\angle DAB = 3(10) + 2(15) = 60°$ **[M1]**

Angle at centre is twice angle at circumference: Obtuse $\angle BOD = 2 \times 60 = 120°$ **[A1]**

[Back to Question] *[Total Marks: 6]*

34 **(a)** Angle $BAD = 180° - 115° = 65°$ **[B1]**

$(2x + 15) + 65 + (x + 10) = 180$ **[M1]**

$3x + 90 = 180$ **[M1]**

30 **[A1]**

(b) Angle $ABD = (x + 10)°$ OR Angle $ADB = (2x + 15)°$ **[M1]**

40 **[A1]**

[Back to Question] *[Total Marks: 6]*

35 **(a)** **(i)** 58 **[B1]**

(ii) Alternate segment theorem **[B1]**

(b) Angle $AQB = 180 - 58 - 58 = 64°$ or Angle $BAQ = 58°$ **[M1]**

128 **[A1]**

(c) Angle $ACB = 180 - 64$ (using cyclic quadrilateral $ACBQ$) **[M1]**

116 **[A1]**

[Back to Question] *[Total Marks: 6]*

36 (a) States Angle CAB = Angle BCQ because of the alternate segment theorem **[B1]**

$2x - 5 = x + 17$ **[M1]**

$x = 22$ **[A1]**

(b) Identifies angle $ACB = 90°$ (angle in a semicircle) or calculates angle $CAB = 39°$ **[M1]**

51 **[A1]**

(c) Angle $ADC = 180 - 51$ (Opposite angles of cyclic quadrilateral sum to 180°) **[M1]**

129 **[A1]**

[Back to Question] *[Total Marks: 7]*

37 (a) Identifies angle ABC = angle CAQ (alternate segment theorem) OR angle ACB = angle PAB **[M1]**

Sets up the equation: $3x - 5 = x + 25$ **[M1]**

$x = 15$ **[A1]**

(b) Calculates angle $ABC = 40°$ and angle $ACB = 40°$ (or uses angles on straight line PQ) **[M1]**

100 **[A1]**

(c) Uses opposite angles of cyclic quadrilateral $ABDC$ sum to 180° $(180 - 100)$ **[M1]**

80 **[A1]**

[Back to Question] *[Total Marks: 7]*

38 (a) Angle TAB = angle ACB (alternate segment theorem) **[M1]**

$2y - 10 = 3y - 40$ **[M1]**

30 **[A1]**

(b) $x + 10 = 180 - 2(50)$ **[M1]**

70 **[A1]**

(c) Angle $ADB = 180 - 50$ (opposite angles of cyclic quadrilateral) **[M1]**

130 **[A1]**

[Back to Question] *[Total Marks: 7]*

39 **(a)** **(i)** Angle ACT = Angle $CAT = \dfrac{180-48}{2} = 66°$ **[M1]**

Angle ABC = Angle $ACT = 66$ (alternate segment theorem) **[A1]**

(ii) Cyclic quadrilateral $ABCD$, so angle $ADC = 180 - 66$ **[M1]**

114 **[A1]**

(iii) $180 - (114 + 31) = 35$ **[B1]**

(iv) Angle at centre $= 2\times$ angle at circumference **[M1]**

$2 \times 66 = 132$ **[A1]**

[Back to Question] *[Total Marks: 7]*

40 **(a)** $180 - 2x$ **[B1]**

(b) $\dfrac{180-2x}{2} = 90 - x$ **[B1]**

(c) Angle $ADB = 180 - (90 - x)$ (opposite angles of cyclic quadrilateral) **[M1]**

$90 + x$ **[A1]**

(d) $90 + x = 3(90 - x)$ **[M1]**

$90 + x = 270 - 3x \implies 4x = 180$ **[M1]**

$x = 45$ **[A1]**

[Back to Question] *[Total Marks: 7]*

41 **(a)** **(i)** $2x + 3y = x + 2y + 10$ (from Alternate Segment Theorem) **[B1]**

$3x - y = x + y + 8$ (from Angles in Same Segment) **[B1]**

(ii) Rearranges to linear equations: $x + y = 10$ and $2x - 2y = 8 \implies x - y = 4$ **[M1]**

Valid method to solve simultaneous equations (e.g. adds equations to get $2x = 14$) **[M1]**

$x = 7$ and $y = 3$ **[A1]**

(b) Uses opposite angles of cyclic quadrilateral: angle $ADC = 180° - 95° = 85°$ **[B1]**

Calculates angle $BDC = (7) + (3) + 8 = 18°$ OR angle $BAC = 18°$ **[B1]**

Finds angle $ADB = 85° - 18° = 67°$ and uses alternate segment theorem
$\implies$ angle $QAB = 67°$ **[B1]**

[Back to Question] *[Total Marks: 8]*

42 **(a)** Sets up equation for x: $2x + 25 = 4x - 5$ **[M1]**

$x = 15$ **[A1]**

Sets up equation for y: $y + 12 = 3y$ **[M1]**

$y = 6$ **[A1]**

(b) Evaluates angles on straight line: $180 - (4(15) - 5) - 3(6)$ OR evaluates angles in triangle
$180 - (2(15) + 25) - (6 + 12)$ **[M1]**

107 **[A1]**

(c) Recognises $ABTC$ is a cyclic quadrilateral and uses $180 - \angle ABT$: $180 - (2(15) + 25)$ **[M1]**

125 **[A1]**

[Back to Question] *[Total Marks: 8]*

43 **(a)** **(i)** Angle $OAT = 90°$ and angle $OBT = 90°$ **[B1]**

Angle $AOB = 180 - x$ **[M1]**

$180 - x = 2(2x - 30)$ **[M1]**

$180 - x = 4x - 60 \implies 5x = 240 \implies x = 48$ **[A1]**

(ii) Angle $AOB = 180 - 48 = 132°$ **[M1]**

Angle $OAB = (180 - 132) \div 2 = 24$ **[A1]**

(b) Angle $ACB = 2(48) - 30 = 66°$ **[M1]**

Angle $ADB = 180 - 66 = 114$ (opposite angles of cyclic quad) **[A1]**

[Back to Question] *[Total Marks: 8]*

44 **(a)** $3x - 11 = 2x + 14$ **[M1]**

$x = 25$ **[A1]**

Theorem: alternate segment theorem **[B1]**

(b) Angle $ABD = 3(25) - 11 = 64°$, so angle $ABC = 64° + 48° = 112°$, giving angle $ADC = 68°$ **[B1]**

Reason: opposite angles of a cyclic quadrilateral sum to 180° (or angle in a semicircle = 90°) **[B1]**

(c) Central angle $AOC = 2 \times 68° = 136°$ **[B1]**

$AC = 2 \times 9.4 \sin 68°$ (or $\sqrt{9.4^2 + 9.4^2 - 2(9.4)(9.4)\cos 136°}$) **[M1]**

17.4 **[A1]**

(d) Central angle $AOD = 2 \times 64° = 128°$ **[B1]**

Area $= \frac{128}{360} \times \pi \times 9.4^2 - \frac{1}{2} \times 9.4^2 \sin 128°$ $(= 98.70 - 34.81)$ **[M1]**

63.9 **[A1]**

[Back to Question] *[Total Marks: 11]*

45 **(a)** Triangle PAB is isosceles $(PA = PB)$ **[M1]**

$\frac{180 - 4x}{2}$ **[M1]**

$90 - 2x$ **[A1]**

(b) Angle $ACB =$ Angle PAB (Alternate segment theorem) **[M1]**

$3x - 10 = 90 - 2x$ **[M1]**

$5x = 100 \implies x = 20$ **[A1]**

(c) $x = 20 \implies$ Angle $APB = 80°$ **[M1]**

Angles in quadrilateral $PAOB$ sum to 360°, with $OAP = OBP = 90°$ **[M1]**

$360 - 90 - 90 - 80 = 100$ **[A1]**

(d) Angle $ACB = 3(20) - 10 = 50°$ **[M1]**

$180 - 50$ (opposite angles of cyclic quad sum to 180°) **[M1]**

130 **[A1]**

[Back to Question] *[Total Marks: 12]*